

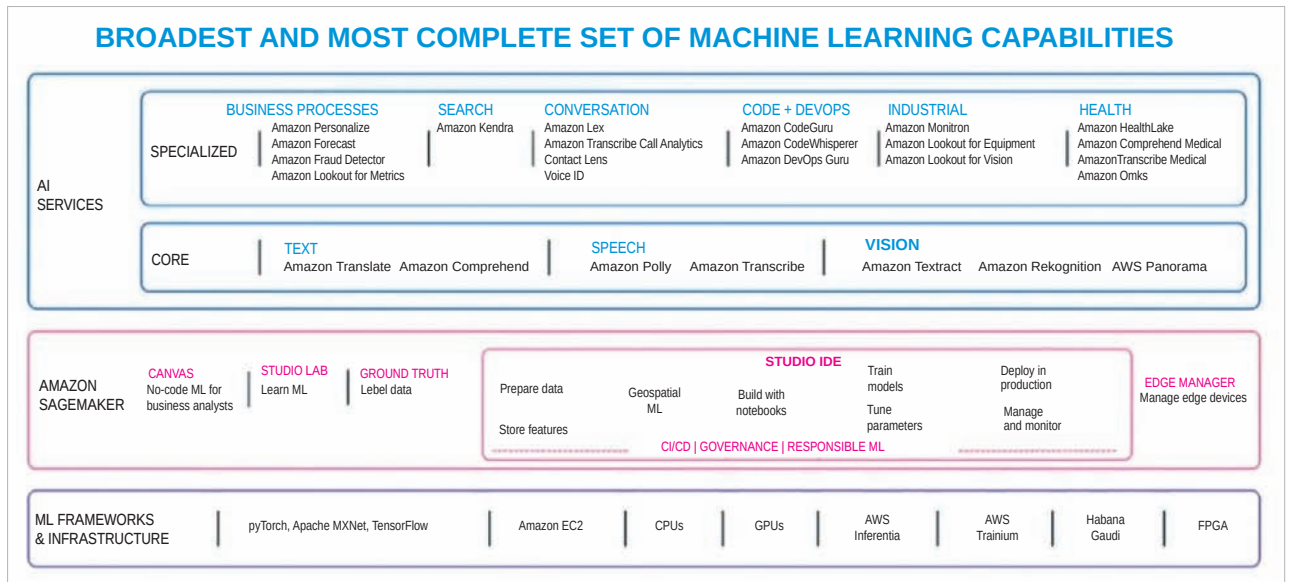


Figure 1: AWS AI/ML stack

For a seamless flight boarding solution, the following services can be used.

Amazon SageMaker: A fully-managed service that provides developers and data scientists with the tools to build, train, and deploy machine learning models at scale.

Amazon Rekognition: A deep learning-based image and video analysis service that can detect objects, faces, and text in images and videos.

Amazon Comprehend: A natural language processing service that can analyse text and extract insights such as sentiment, entities, and key phrases.

Amazon Polly: A text-to-speech service that can convert text into lifelike speech in multiple languages and voices.

Amazon Translate: A neural machine translation service that can translate text between languages with high accuracy.

Amazon Transcribe: An automatic speech recognition service that can transcribe audio and video files into text.

Amazon Forecast: A fully managed forecasting service that uses machine learning to deliver highly accurate forecasts.

Amazon Personalize: A service that provides real-time recommendations

for content, products, and services using machine learning algorithms.

Amazon SageMaker

Amazon SageMaker is a powerful machine learning platform with a standard interface that provides a complete set of tools and services to build, train, and deploy machine learning models at scale quickly. SageMaker uses containers to wrap favourite algorithms and frameworks, including built-in algorithms like XGBoost, DeepAR, and FM, as well as frameworks like PyTorch, SKLearn, and TensorFlow.

Some of the key services provided by Amazon SageMaker include:

Data labelling: SageMaker's Ground Truth provides a fully managed data labelling service that makes it easy to label data sets using human annotators or pre-built machine learning models.

Model building: SageMaker provides a range of built-in algorithms and frameworks for building, training, and deploying machine learning models. It supports custom algorithm development using popular frameworks like TensorFlow, MXNet, and PyTorch.

Model training: SageMaker provides a scalable and distributed training environment (using GPUs and multiple instances) that helps train efficient

machine learning models quickly.

Model hosting: It provides a fully managed model hosting environment enabling developers to deploy machine learning models as APIs with auto-scaling, monitoring, and debugging capabilities.

Model tuning: Amazon SageMaker provides an automatic model tuning service that enables data scientists to optimise hyperparameters and improve model accuracy without manual intervention.

Real-time inference: It provides a fully managed, highly available real-time inference service that can scale to handle millions of requests per second.

Batch inference: Amazon SageMaker provides a fully managed batch inference service that can process large volumes of data and give predictions in a cost-effective manner.

End-to-end ML workflow: It provides an end-to-end machine learning workflow that includes data preparation, feature engineering, model training, deployment, and monitoring.

Integration with other AWS services: Amazon SageMaker integrates with other AWS services like S3, Lambda, Step Functions, and CloudFormation to provide a seamless machine learning experience.